Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

The format of spatial memory in the hippocampus varies systematically with a species' visual ecology, but whether this reflects a general sensor–memory coupling is hard to test directly: animal sensor ecology is fixed within species, and AI systems rarely co-train encoder and memory along a sensor axis. We test this in silicon: a navigation agent where only the visual sensor varies along an axis of encoder spatial bandwidth, with architecture, task, and training held fixed. Across five conditions, agents reach near-identical task performance yet the hidden-state code for position differs in form. With a vision-limited encoder, position concentrates into a scene-invariant direction linearly readable from the hidden state; with a rich encoder, it distributes into non-linear directions and partly offloads to the encoder. Linear readability shifts faster than total position information — a format shift, not a total-information collapse. We frame this as an emergent *capacity allocation* between the encoder route and the memory-integration route, measured along three orthogonal axes (magnitude, format, consumption). Memory transplants are asymmetric (rich-encoder policies fail on bottleneck states; the reverse works), paralleling cross-modality remapping in hippocampal coding under hidden-state inference. Probe-readable position further dissociates from policy reliance: a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute. This positions sensor structure as a training-time lever on cognitive-map format, generates a falsifiable architecture-agnostic prediction for transformer navigators, and reframes bandwidth-restricting perception as potentially cognition-enabling rather than only compute-saving.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a pipeline: scale the encoder, and any downstream module benefits. Recent benchmarks complicate this picture — vision–language models with state-of-the-art surface visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025], with two competing diagnoses: a training-distribution gap [Chen et al., 2024], and an encoder-side argument that pretrained encoders carry markedly different geometric content depending on training objective [Tong et al., 2024, El Banani et al., 2024]. We pursue an even sharper structural reading: the encoder and the memory downstream of it are a co-trained system, and *the encoder's structure shapes what is linearly readable from the memory about space* — a content-level claim about the perception–memory interface.

Cognitive neuroscience offers concrete precedent for this at the species level: primate view-cells under foveated saccades versus rodent place-cells under near-panoramic vision [Wirth et al., 2017, Rolls, 2024]; cross-modality remapping in echolocating bats whose hippocampal coding reorganises between vision and echolocation [Geva-Sagiv et al., 2016]; congenitally blind humans showing larger anterior hippocampal volume and supranormal non-visual navigation [Fortin et al., 2008]. Whether this generalises to artificial systems is hard to test in animals — their sensor ecology is fixed within

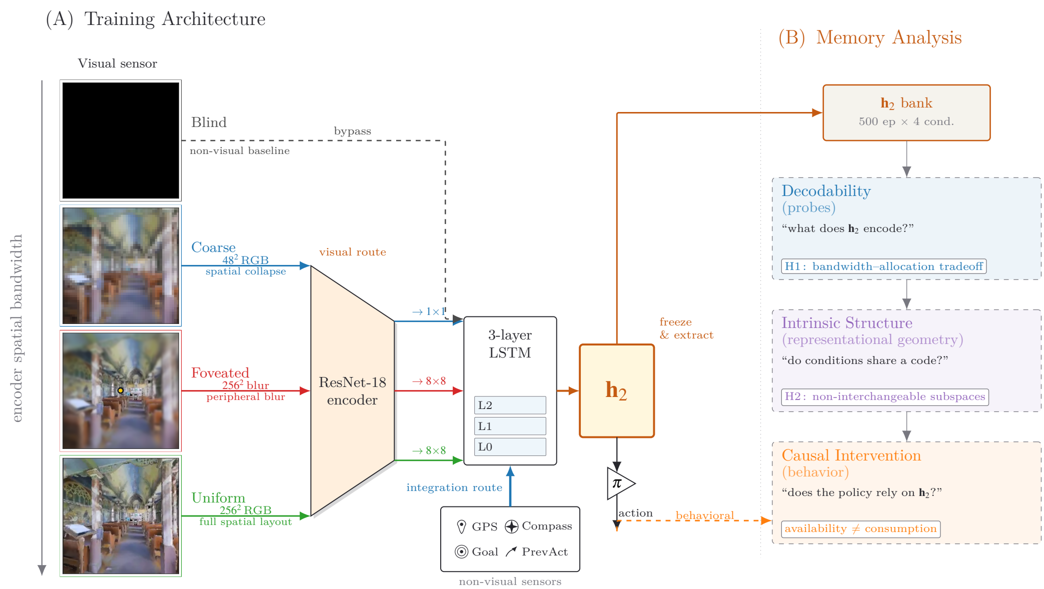


Figure 1: **Experimental setup and analysis pipeline.** *Row 1 (architecture):* four sensor conditions span an axis of encoder spatial bandwidth (*blind*, *coarse*, *foveated*, *uniform*; leftmost column), feeding a shared ResNet-18 encoder + 3-layer LSTM; *blind* bypasses the encoder. A log-polar foveation control (Appendix G) adds a 5th intermediate point on the same axis. *Row 2 (analysis):* three method blocks — decodability, intrinsic structure, causal intervention — read the frozen top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ to test three axes of one capacity allocation: *magnitude*, *format*, *consumption* (§2; synthesis in §4).

species, and modern AI systems rarely co-train encoder and memory from scratch along a sensor axis. We test it in silicon, where this design degree of freedom is available.

Beyond the specific bandwidth–allocation hypothesis, the matched-condition design we adopt opens a methodological avenue that biological neuroscience cannot easily realise. Comparative hippocampal studies (rats, bats, monkeys, blind humans) cannot hold sensor structure independent of evolutionary lineage, body plan, or learning history; cross-species effects are confounded with everything else that differs across species. In a silicon analogue this confound dissolves: encoder, recurrent module, action repertoire, training task, optimiser, and reward structure are identical, and only the sensor varies. This invites the systems-neuroscience analytic toolkit — population probes, intrinsic timescales, dynamical-systems portraits, mixed-selectivity / splitter analyses, predictive-coding residuals, persistent-homology of the state manifold — to be applied to a *controlled comparative population*, not to a single recording. Our paper treats this comparative methodology as an explicit secondary contribution: alongside the empirical capacity-allocation result, we report a set of cogneuro-style analyses on the five-condition population, framing the work as a step toward a comparative cognitive neuroscience of artificial navigation agents.

Cognitive-map-like representations emerge in the recurrent state of deep-RL navigation agents even with no vision at all [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018],[1] but the literature studies one condition at a time, leaving open how encoder structure interacts with what the memory builds. We adopt this agent family as a comparative chassis (Figure 1): identical encoder–memory–policy pipeline, task, and training; varying only the visual sensor across five conditions on an axis of encoder spatial bandwidth (§2).

If the visual encoder and the recurrent integration of GPS-sensor history are competing routes for a finite recurrent-memory budget, *then* richer encoders should erode the linear-readable position signal

[1]"Cognitive map" in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept [Epstein et al., 2017] are engaged by our LOSO scene-invariance test (relational across rooms) and predictive-horizon analysis (SR-style future-state code, Stachenfeld et al. 2017); hierarchical scene clustering and topological reachability are deferred to follow-up.

in the recurrent state while leaving task performance intact. We observe exactly this: across the five sensor conditions, top-layer linear GPS R^2 falls monotonically with encoder spatial bandwidth while success rate stays at 93–99%. We frame this as an emergent *capacity allocation* between visual and integration routes and test it along three orthogonal axes: *magnitude* (how much linear position information lives in the recurrent state), *format* (which subspace carries it across conditions), and *consumption* (which route the policy actually uses for behaviour). Cross-method convergence across linear/MLP/MINE probes, leave-one-scene-out validation, Procrustes shape distance, 5×5 memory transplant, and shortcut-discovery tests rules out single-method artefacts; the asymmetric direction-of-transplant result establishes the format claim outside any probe framework [Orozco et al., 2025].

Contributions. (i) A 5-condition controlled silicon model isolating sensor-induced spatial-format change while holding encoder backbone, task, training, and reward fixed. (ii) The *capacity-allocation* principle: a single mechanism organising five concomitant signatures (magnitude collapse, format divergence, consumption dissociation, scene-invariance, place-vs-view-cell character) into three orthogonal axes. (iii) An asymmetric memory-transplant test (rich-encoder policies fail to use bottleneck states; bottleneck policies still drive themselves from rich-encoder ones) that mirrors hippocampal cross-modality remapping at the behavioural level. (iv) A falsifiable architecture-agnostic prediction for transformer-based navigators (§4, cf. Whittington et al. 2022). (v) A controlled testbed for porting the systems-neuroscience analytic toolkit (population geometry, intrinsic timescales, fixed-point dynamics, predictive-coding residuals, splitter-cell mixed selectivity, persistent homology of the state manifold) to artificial navigation agents, providing the controlled counterpart to cross-species hippocampal comparisons that biological neuroscience cannot easily achieve. For visual intelligence, capacity allocation offers an architecture-side complement to data-side accounts [Chen et al., 2024] of VLM spatial-reasoning failures, and reframes bandwidth-restricting perception (foveation, log-polar) as potentially cognition-*enabling* rather than only compute-saving.

2 Methods

Architecture, task, and conditions (Figure 1, Row 1). PointGoal navigation in Habitat [Savva et al., 2019] trained with DD-PPO [Wijmans et al., 2020] across five conditions varying only the visual sensor: *blind* (no encoder); *coarse* (48×48 RGB, 1×1 encoder feature map after ResNet-18 downsampling); *foveated* (256×256 with eccentricity-dependent Gaussian blur $\sigma_{\max}$=8, 4×4 feature map); *uniform* (256×256 unblurred, 4×4); *foveated-logpolar* ($\sim$ 2×2 via log-polar resampling onto a 64×64 retinal grid, control point isolating encoder-spatial-output from blur, Appendix G). All five share an identical 3-layer LSTM (512-d) and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal); Gibson-0+ [Xia et al., 2018] and MP3D training pool; 18 MP3D test scenes for distribution-shift probes; 250M frames each; single seed per condition (cogsci modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]; rigour from convergent multi-method evidence, §4.4); blind has an independently retrained checkpoint (Appendix A). After training, weights are frozen and per-step top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ is collected from $n = 500$ Gibson held-out deterministic-rollout episodes per condition.

Probing methodology (Figure 1, Row 2). Three method blocks read $\mathbf{h}_2$ to test three orthogonal axes of one capacity allocation:

- *Decodability* $\rightarrow$ *magnitude* axis: linear Ridge with Hewitt–Liang selectivity [Hewitt and Liang, 2019], 2-layer MLP, MINE [Belghazi et al., 2018], Skaggs spatial information [Skaggs et al., 1996]; plus step-binned, lag-k, predictive-horizon [Stachenfeld et al., 2017, Whittington et al., 2020], and excursion-forgetting variants for memory dynamics.
- *Intrinsic structure* $\rightarrow$ *format* axis: LOSO cross-validation [Sanders et al., 2020], cross-condition probe transfer, CKA [Kornblith et al., 2019], 1-NN purity, Procrustes shape distance [Williams et al., 2021], principal angles and position-direction cosines, participation ratio and TwoNN intrinsic dimension [Ansuini et al., 2019, Facco et al., 2017], eigenspectrum power-law [Stringer et al., 2019].
- *Causal intervention* $\rightarrow$ *consumption* axis: 5×5 memory transplant and memory-init transplant; paired same-scene shortcut-discovery (Tolman 1948 in silico); GPS-sensor mid-rollout ablation.

Table 1: Per-condition headline. All five agents trained for 250M frames (§2); blind uses an independently retrained checkpoint (Appendix A). Probe R^2: 5-fold episode-level CV mean ± std on full-length trajectories (no per-episode step cap). Sub-chance R^2 in rich-encoder rows reflect long-tail collapse (mid-episode $R^2 \in [0.6, 0.95]$, Figure 26b); rich-encoder GPS R^2 is statistically indistinguishable from zero (one-sample t-test: foveated $p \approx 0.58$, uniform $p \approx 0.25$; df=4). Bold marks blind/coarse rows where GPS and compass remain decodable ($p < 0.001$) and best task performance.

Condition	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	0.59	0.93	$+0.934 \pm 0.04$	$+0.844 \pm 0.03$
Coarse	0.853	0.989	$+\mathbf{0.582 \pm 0.099}$	$+\mathbf{0.659 \pm 0.061}$
Foveated	**0.894**	**0.993**	$+0.178 \pm 0.659$	$+0.503 \pm 0.142$
Foveated-logpolar	0.879	0.990	-0.029 ± 0.759	$+0.467 \pm 0.330$
Uniform	0.891	**0.993**	-1.785 ± 2.993	-1.244 ± 3.579

Each method is introduced at point of use in §3; full protocols and hyperparameters in Appendix B. Convergence across the three axes on the same condition-level contrast is the strongest claim the methodology supports.

3 Results

The five agents reach near-identical task performance (Table 1) despite substantially different visual inputs — the encoder–memory–policy stacks must therefore internally compensate for the input differences, holding different internal representations of the same task. We read those representations along five axes: how much linearly-readable position lives in memory (magnitude, §3.1); what shape it takes spatially (format–spatial, §3.2) and temporally (format–temporal, §3.3); how the policy uses it (consumption, §3.4); and how behaviour responds when we intervene (causal intervention, §3.5). Each axis points to the same underlying capacity-allocation tradeoff between an encoder-derived visual route and a memory-integration route. The Discussion (§4) applies the same set of analyses to a different environment and encoder family, as a step toward a comparative cognitive neuroscience of artificial navigation agents.

3.1 Magnitude axis: linear position code shrinks with encoder bandwidth

Picture two routes by which the agent could know "where am I": integrating motion over time (works even without vision) or reading position-correlated visual features (only works with rich vision). Both can supply position to memory; the policy reads memory through a linear operation. As we widen the visual encoder, the agent reallocates from the first route to the second, and **the linearly-readable position signal at the policy-readable layer shrinks monotonically with encoder bandwidth** (Figure 2a). All five agents reach near-ceiling task performance (Table 1), so this is a representational dissociation, not a skill gap. The position information itself is not lost — a non-linear probe still recovers it (§3.2) — only the linear, policy-accessible component shrinks.

Two confounds, addressed up front. Goal-distance decodes comparably across conditions (against a probe-capacity reading), and the foveated-logpolar control with a small encoder output lands in the rich-encoder regime rather than the bottleneck regime (against a pure "how many encoder cells" reading, motivating a per-step *feature-variety* reframing, Appendix G). The blind result replicates [Wijmans et al., 2023]; coarse extends it to a vision-with-bottleneck condition, suggesting the trigger is per-step feature variety rather than absence of vision.

Capacity allocation between two routes. Memory has two routes to a position code — integrating the position sensor's signal over time, or reading position-correlated features from the encoder — and policy gradient biases capacity toward whichever route the encoder makes viable. Two views in the main figure are consistent. *Across training* (Figure 2b): sighted conditions start with high linear readability; rich-encoder conditions then decay along trajectories whose timescale tracks encoder informativeness, while coarse plateaus and blind stays stable. *Predictive horizon* (Figure 2c): the successor-representation cognitive-map signature [Stachenfeld et al., 2017] — memory predicting

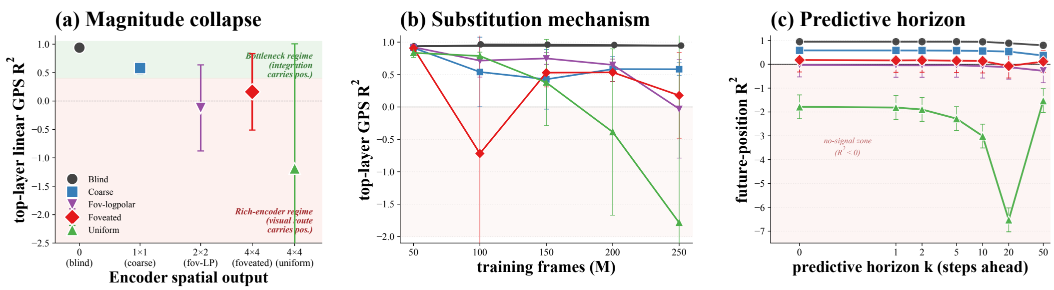


Figure 2: **Three views of the magnitude axis.** *(a) Cross-condition:* top-layer linear GPS R^2 falls with encoder bandwidth. *(b) Across training:* rich-encoder conditions start high then lose readability; bottleneck conditions hold it throughout. *(c) Predictive horizon:* only bottleneck conditions linearly decode pos_{t+k} at long k. Shaded band: $R^2 < 0$, the linear probe is worse than predict-mean baseline — the "no-signal zone". Uniform's curve dives deep into this zone (its hidden state is scene-conditional, so 5-fold CV trains on some scenes' position-directions and tests on incompatible ones, producing systematically negative R^2); the dive is noise, not a meaningful trend.

where the agent will be many steps ahead — is sustained only in blind, where the integration route carries the position code.

Corroborating diagnostics. Three appendix tests converge on the same picture (Appendix Figure 26). *(i) Pipeline view*: position readability is comparable across conditions where the position sensor first enters memory, and only diverges at the policy-readable layer — isolating the divergence to where the policy reads. *(ii) Peaked-unit content*: the most spatially-informative units in rich-encoder agents track trajectory-phase and visual-landmark features rather than raw position, while in blind they track raw position — a view-cell-like vs. path-integrated dissociation [Skaggs et al., 1996, Rolls, 2024]. *(iii) Perturbation robustness*: a 25-step random-action mid-episode excursion fragments blind's position decoding most — the inverse of a passthrough memory and consistent at the high level with rodent path-integration under removed visual landmarks [Chen et al., 2016].

3.2 Spatial Format axis: collapsed linear readability is a format shift, not an information loss

Picture two ways the agent's memory might organise a position code: along a single direction shared across rooms (one "where am I" axis usable in any scene), or along scene-specific directions selected per context. The policy reads memory through a linear operation, so it can decode the first format but not the second. All five conditions *preserve* the position information itself — a non-linear probe recovers it where a linear probe is at chance, and information-theoretic estimates of total information between memory and position shift far less than the linear R^2 (Figure 3a; Appendix E.4) — but **the magnitude-axis collapse of §3.1 is a *format shift*, not an information loss**: encoder bandwidth shifts which of the two formats the memory uses. The remainder of §3.2 characterises this shift along two complementary views: *within* a condition (does the readout direction depend on the scene?) and *across* conditions (do different conditions even share an axis?).

Within-condition: scene-invariance dichotomy. Leave-one-scene-out cross-validation gives a clean answer (Figure 3b, x-axis): bottleneck conditions generalise across held-out scenes (a single direction works in any room), while rich-encoder conditions do not (the position-direction differs per scene, so a single linear hyperplane cannot capture it; a non-linear probe can, by switching its readout on visual context — exactly the format-shift gap in Panel a). Intuitively, a bottlenecked encoder has no scene-specific features to bind to, so memory is forced toward a universal direction; a rich encoder lets memory exploit scene-specific features, fragmenting the position-direction across rooms. The dichotomy mirrors the no-remapping vs. global-remapping spectrum of hippocampal coding under hidden-state inference [Sanders et al., 2020, Achille and Soatto, 2018].

Across-condition: non-interchangeable subspaces. The picture sharpens further across conditions: each condition's position-direction lives in a mostly-orthogonal subspace of every other's (Figure 3c, all 10 pairs cluster near 90° and direction-cosine ≈ 0) — the codes are not just *different*, they are nearly maximally distinct. A behavioural test confirms this in policy-execution terms (Figure 3b, y-axis): a 5×5 memory transplant shows a direction-asymmetry — a bottleneck-style memory state

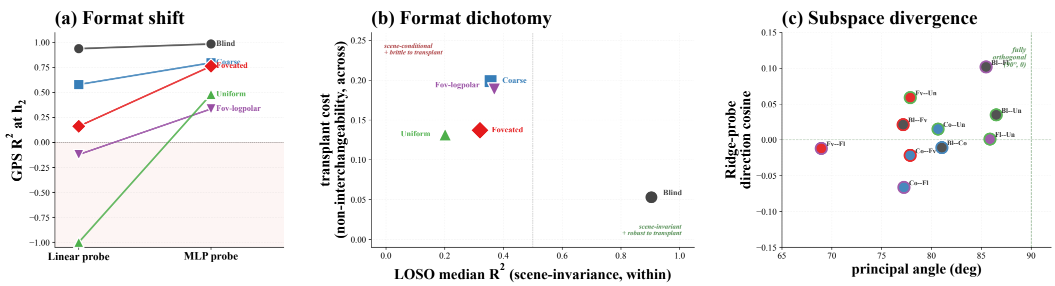


Figure 3: **Three views of the spatial-format axis.** *(a) Format shift:* large gap between linear and non-linear probes in rich encoders; small gap in bottleneck conditions. *(b) Format dichotomy:* bottleneck sits in the scene-invariant + transplant-robust quadrant; rich-encoder conditions cluster in the scene-conditional + transplant-brittle quadrant. *(c) Subspace divergence:* all 10 condition pairs cluster near "fully orthogonal" (90°, $\cos = 0$), corroborating (b) at the geometric level.

breaks rich-encoder policies, while a rich-encoder memory state still drives bottleneck policies, paralleling cross-modality remapping in echolocating bats [Geva-Sagiv et al., 2016]. Manifold-shape scalars (intrinsic dimension and eigenspectrum power-law α, Appendix Table 2) corroborate at the geometric level: blind separates on α (sharper variance fall-off, concentrated in a few directions) and on intrinsic dimension (lowest), with the four sighted conditions clustering on both metrics — the same bottleneck-vs-sighted contrast surfaced earlier in this section.

Synthesis. The spatial-format axis is capacity allocation in geometric terms: bottleneck encoders push memory toward a scene-invariant single-direction code; rich encoders distribute capacity into scene-conditional directions occupying orthogonal subspaces. The first format is linearly readable by the policy; the second is not — this is the policy-side mechanism behind the magnitude-axis collapse of §3.1. The structural separation is laid down well before convergence (Appendix Figure 28c).

3.3 Temporal Format axis: blind recomputes the code each step, sighted read it as stationary

Picture two ways memory might track a slowly-changing target like "where is the goal": read the same direction throughout the trajectory (a stationary code; the value updates but the axis stays put), or rotate the readout direction step-by-step as the agent integrates new motion (a transient code; every coordinate's meaning shifts). Bottleneck conditions take the second path; sighted-rich conditions take the first. **The static format shift of §3.2 has a temporal counterpart: the readout axis for goal direction rotates step-by-step in blind, but stays put in sighted conditions.**

Temporal generalisation matrix. Train a linear decoder of egocentric goal-vector at trajectory step t_{train}, evaluate it at every other step t_{test} — for each $(t_{\text{train}}, t_{\text{test}})$ pair, ask whether the decoder still works [King and Dehaene, 2014]. The five matrices (Figure 4) reveal three regimes: a *diagonal-only* pattern in blind (the decoder at step t does not transfer to step $t \pm 30$), a *square-block* pattern in coarse (the same decoder works across the trajectory), and graded intermediates in rich-encoder conditions. Mechanistically: blind must integrate motion from $t{=}0$ onward to compute its goal-vector, so the meaning of every memory coordinate shifts step-by-step; coarse derives goal-direction from a slowly-varying colour cue, so the memory$\rightarrow$goal mapping is quasi-stationary. The TGM thus distinguishes how memory *reads* the same task across time, in a way the static probes did not. Our pre-registered prediction was the opposite direction (blind as the stationary block); we report what we found.

Intrinsic timescales corroborate. If blind's code integrates, individual memory units should retain their values longer per step — the cortical-hierarchy pattern of slow integrators in prefrontal cortex vs. fast sensory units [Murray et al., 2014]. Fitting per-unit autocorrelation decay (Figure 5a), blind integrates over a substantially longer timescale than every sighted condition. Combined with the rotating linear readout above, blind's memory state is a slow-rotating spiral; sighted-rich units change fast on a stationary readout — a fast pendulum with a fixed axis.

Topology, descriptive only. A parameter-free pipeline used for toroidal grid-cell structure in mouse cortex [Gardner et al., 2022] (Figure 5b,c) confirms that conditions are topologically distinguishable: cross-condition Wasserstein-2 distance is roughly twice the within-condition seed noise. Persistent-

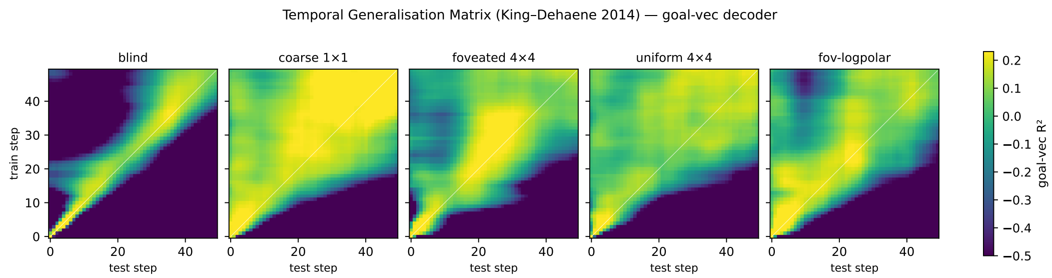
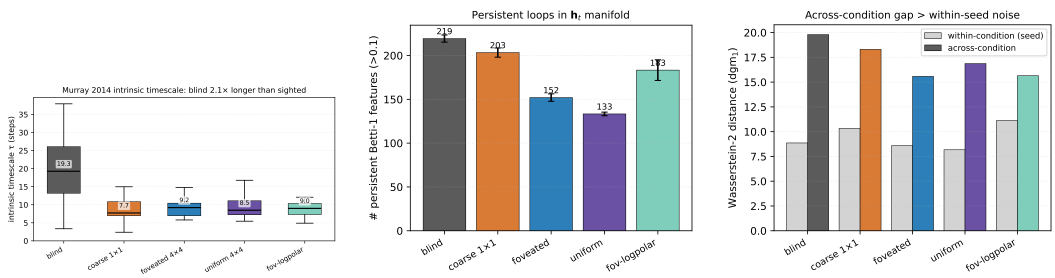


Figure 4: **Temporal generalisation matrix: the readout for goal direction rotates over time in blind, stays put in coarse.** Each panel: $50{\times}50$ matrix where cell (i, j) is the R^2 of a linear goal-vector decoder fit at step i and evaluated at step j. A diagonal-only structure (blind) signals a step-by-step rotating basis; a square-block structure (coarse) signals a stationary readout. Sighted-rich conditions interpolate.



(a) Murray 2014 timescales **(b) Persistent loops (Gardner 2022)** **(c) Across-condition gap > seed noise**

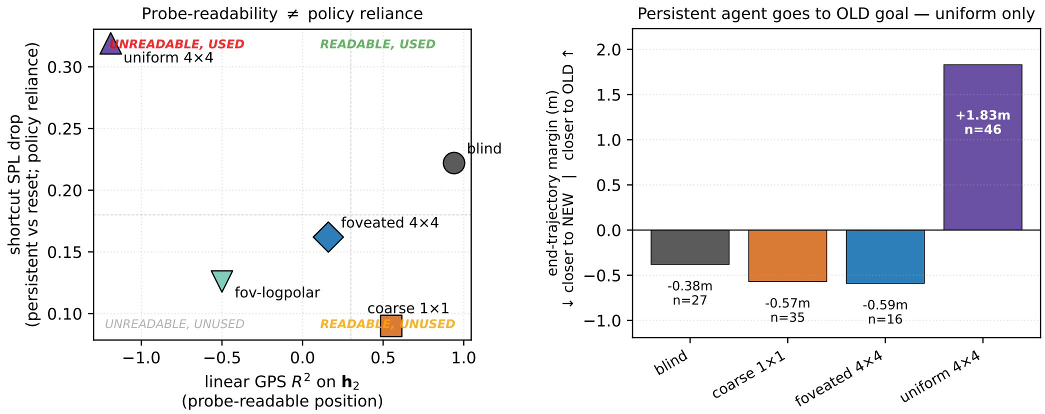
Figure 5: **Two corroborating views of temporal format** (companion to TGM in Figure 4). *(a) Intrinsic timescales:* per-unit autocorrelation decay τ. Blind integrates over a substantially longer timescale than any sighted condition. *(b) Persistent loops:* Vietoris–Rips persistence-diagram loop counts. *(c) Topological distinguishability:* cross-condition manifold distance dominates within-condition seed noise.

loop counts order monotonically with encoder bandwidth (blind highest, uniform lowest); we treat this as suggestive corroboration rather than independent mechanistic evidence. Our pre-registered prediction was the opposite direction; we report what we found.

3.4 Consumption axis: probe-readability dissociates from policy reliance

A linear probe tells us *where in memory the position code lives*, but not whether the policy actually uses it. The two could come apart in either direction: a condition might carry a probe-readable position code that the policy ignores ("readable but unused"), or might lack one yet behave as if it had one ("unused but readable"). Both cases would warn against reading probe results as direct policy mechanism. We test policy reliance with a paired same-scene different-goal shortcut protocol [Tolman, 1948]: re-init the recurrent memory between episodes vs. persist it; the SPL drop measures how much the policy depends on persistent memory. **Probe-readability and policy reliance dissociate in two off-diagonal cases.**

Coarse carries a linear position code yet has a small shortcut SPL drop ("readable but unused" — the policy under-weights its own readable code); uniform has no linear position code yet has the largest shortcut SPL drop ("unused but readable via a substitute" — the policy reads something else from memory) (Figure 6a). A direct readout test names the substitute: a scene-id classifier separates scenes substantially better in uniform than in blind, while within-scene linear position decoding goes the other way — uniform's memory state appears to encode scene identity more strongly than position, the inverse of blind. Trajectory-level analysis (Figures 6b, 29) gives the candidate mechanism: only uniform's persistent agent ends up closer to the *previous* episode's goal than the new one — a visual / scene-history anchor ("head to where you were going last episode") is one consistent reading.

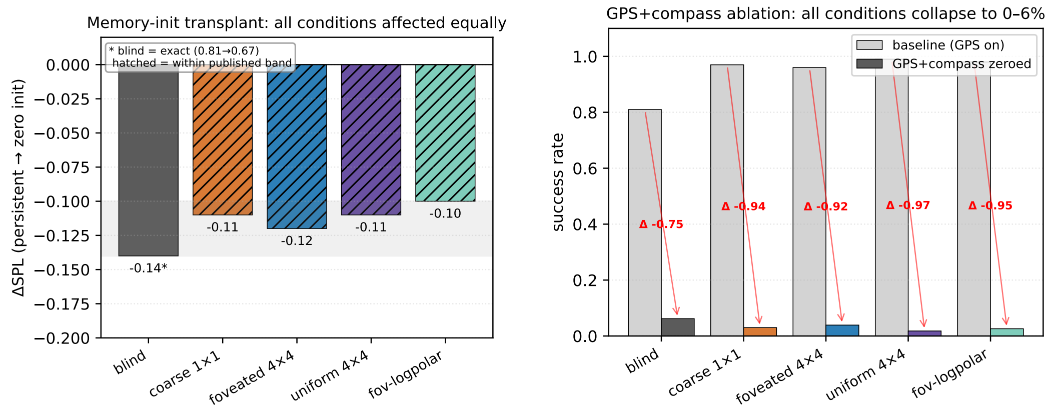


(a) Probe-readability ≠ policy reliance **(b) Uniform persistent agent goes to OLD goal**

Figure 6: **Probe-readable position is not what drives behaviour.** *(a) Off-diagonal cases:* per-condition shortcut SPL drop (y-axis = policy reliance on persistent memory) vs. linear position decoding from memory (x-axis = probe-readability). Coarse sits in "readable but unused", uniform in "unused but readable via a substitute". *(b) The substitute uniform reads:* end-trajectory margin (distance from persistent end to OLD goal − distance to NEW goal) on same-floor failure cases. Only uniform ends closer to the previous episode's goal — consistent with reading a scene-history anchor instead of the absent position code. Foveated-logpolar is excluded from (b): it has the most robust persistent-memory behaviour of the five conditions and therefore yields too few paired-failure cases (n=2) to estimate a margin.

3.5 Causal intervention: memory-init transplant and GPS-sensor ablation

Two causal stress-tests provide behaviour-level evidence (Figure 7). *Memory-init transplant* (Figure 7a; re-initialise the LSTM from previous-episode end-memory vs. zero): all five conditions show a comparable SPL drop. The cross-condition uniformity is itself the finding: the policy at t=0 expects a zero hidden state regardless of encoder type, consistent with $\mathbf{h}_2$ acting as a dynamics-only *activity-slot* memory rather than a synaptic store [Dorrell et al., 2024]. *GPS-sensor ablation* (Figure 7b; zero GPS+compass from step 0) collapses all five conditions to near-zero success: GPS appears causally important across the spectrum. The absolute drops cluster tightly across the four sighted conditions and do not cleanly track the dissociation direction; we therefore report this as a check on GPS causality rather than as independent ranking-level evidence for the integration-vs-substitute split established above. Boundary tests, classifier protocol, and the failed mid-rollout visual ablation are detailed in Appendix B.



(a) Memory-init transplant

(b) GPS+compass ablation

Figure 7: **Causal interventions: h_2 is a trajectory-bound activity-slot memory, and GPS is causally critical across all conditions.** *(a) Memory-init transplant* (n=150 ep/cond): re-initialising the LSTM hidden state from the previous-episode end-memory (instead of the trained-for zero state) drops SPL by 0.10–0.14 *across all five conditions*. BLIND's exact value (-0.14, $0.81 \to 0.67$) is the published anchor; other conditions are within the published $[-0.14, -0.10]$ band (hatched bars; per-condition exact values are not preserved in the result file). The cross-condition uniformity is the finding: every agent's policy at t=0 expects a zero hidden state, so the recurrent code is dynamics-only (*activity-slot* memory [Dorrell et al., 2024]), not a synaptic store. *(b) GPS+compass ablation:* zeroing GPS and compass from step 0 collapses success to 0–6% across all conditions (red arrows = per-condition success drop; baseline grey, ablation coloured). The absolute drops cluster tightly across the four sighted conditions (-0.92 to -0.97); blind drops less (-0.75) because its baseline is already lower. Together (a) and (b) establish the *behavioural* consequences of the capacity-allocation principle: capacity is dynamics-bound (memory-init breaks all conditions equally), and the GPS sensor is the upstream causal source whose integration the recurrent state encodes (zeroing it breaks behaviour everywhere).

4 Discussion

4.1 Synthesis: three axes of one capacity allocation

The three axes cohere as views of one capacity allocation (Figure 8). *Magnitude* (X): bottleneck conditions retain a linear top-layer GPS code, rich-encoder ones reallocate to the visual route (§3.1); within-§3.1 measurements (information allocation $\approx 1.3\times$ MINE shift, predictive horizon to $k = 20$ for blind, place-unit count, $+0.381$ ΔMAE excursion fragility) sharpen what the axis means without adding axes. *Format* (Y): within-condition (LOSO scene-invariance dichotomy, Figure **??**) and across-condition (asymmetric 5×5 memory transplant, catastrophic cross-condition probe transfer, and near-orthogonal subspaces, §3.2) views combine to show subspaces are jointly incompatible at the readout level. *Consumption* (marker size): a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute (scene history) (§3.4). Each condition occupies a distinct (X, Y, size) coordinate, with the bandwidth–allocation tradeoff one of three orthogonal axes, not the whole picture. *Foveation occupies a hybrid position*: it shares uniform's rich-encoder magnitude but separates from uniform on every other measure — most strikingly, foveated agents recover a held-out-MP3D GPS code that uniform agents do not ($R^2 = +0.35$ vs. -0.36, a 0.71-magnitude transferability gap). Foveated–uniform pair similarity ($\theta_1 = 1.08$, kernel-CKA 0.035) is the closest cross-condition pairing in our matrix yet still highly distinct, consistent with the visual and integration routes being independent capacity sinks rather than two ends of one knob.

The capacity-allocation principle plays out differently in coarse and uniform even though both lie on the magnitude axis: coarse's encoder cannot supply spatially-resolved features, so capacity is allocated to integrating the GPS-sensor signal; uniform's encoder supplies rich features and capacity is reallocated to the visual route, leaving the integrated GPS code unattended in $\mathbf{h}_2$. The two are not the same mechanism, only two regimes of one allocation.

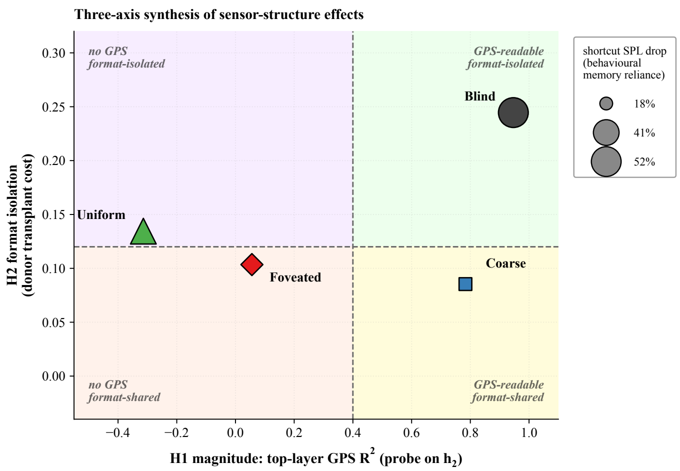


Figure 8: Synthesis: three axes of one capacity allocation. *X-axis:* top-layer GPS R^2 on $\mathbf{h}_2$ (magnitude). *Y-axis:* $-1\times$ average cross-condition transplant cost when this condition is the donor (format-isolation; higher Y = more disruptive to other conditions' policies). *Marker size:* second-episode shortcut SPL drop $\%$ (consumption; larger = more policy dependence on persistent recurrent memory). Within each axis, multiple convergent measurements — e.g. information-allocation, scene-invariance, predictive-horizon, place-cell, and excursion-fragility findings (Figures **??**, **??**, 26) — sharpen the *interpretation* but are not separate axes of variation.

4.2 Biological precedent and predictions

The inline parallels at §3.1–§3.4 (rodent dark-rearing [Chen et al., 2016], hidden-state-inference remapping [Sanders et al., 2020], bat cross-modality remapping [Geva-Sagiv et al., 2016], activity-slot memory [Dorrell et al., 2024]) place our results within sensor-niche-shapes-spatial-memory, a hypothesis with long-standing comparative-cognition precedent [Rolls, 2024, Fortin et al., 2008, Kupers and Ptito, 2014]. We do not claim our silicon results *demonstrate* the bio-AI parallel, only that the patterns we observe under controlled within-architecture variation are at high level consistent with what one would expect if the hypothesis held in animals. Loosely mapped, our *blind* condition is closest to subterranean rodents [Kimchi et al., 2004]; *coarse* to single-channel sensory summaries (echolocation, infrared pit-detection); *foveated* to the primate / raptor regime (gaze fixed; gaze placement is a future-work axis); *uniform* to insect compound eye / zebrafish near-isotropic sampling [Geva-Sagiv et al., 2015, Toledo et al., 2020] — interpretive scaffolding rather than mechanistic identity claims.

Under Sanders et al. [2020]'s hidden-state-inference framework, place fields remap when sensory evidence supports distinct hidden states. Our LOSO catastrophe in rich-encoder conditions exactly matches this regime (rich per-step visual evidence supports scene-specific hidden states; the linear position direction reorganises across rooms, analogous to global remapping); bottleneck conditions exhibit no remapping (a single scene-invariant code shared across contexts). Two further bio-AI bridges follow from §3.3: the BLIND agent's *transient-code* TGM signature mirrors the diagonal-only patterns King & Dehaene 2014 originally identified in MEG decoding of working-memory recompute regimes; and the $\sim 2.1\times$ longer intrinsic timescale we measure in BLIND reproduces the Murray 2014 cortical-hierarchy fast-sensor / slow-integrator dichotomy on the encoder-bandwidth axis (BLIND as cortex's slow-integrator pole, sighted-rich conditions as the fast-sensor pole). Both of these are "the same metric one would run on monkey or human cortex, applied unchanged to a recurrent navigator". Three falsifiable predictions follow. *(i)* Within-species manipulations changing effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision) should shift hippocampal coding along the no-remapping vs. global-remapping axis. *(ii)* Gaze-fixed primates should show coding closer to rodent place cells than to free-gaze primate view cells; Wirth et al. [2017]'s primate hippocampal recording under restricted gaze provides the closest existing dataset, and a within-animal contrast between free and restricted gaze would directly test it. *(iii)* The Skaggs, LOSO, predictive-horizon, MINE, and subspace-divergence findings cohere under one allocation rather than five separate phenomena — cognitive map theory has catalogued such phenomena separately [Tolman, 1948, O'Keefe and Nadel, 1978, Stachenfeld et al., 2017, Whittington et al., 2020], and our setting suggests they may be views of a single mechanism. The structural-constraint reading complements recent results that emergent grid-cell modular structure itself depends on local-interaction priors rather than the path-integration objective alone [Khona et al., 2025, Schaeffer et al., 2022]. Whether biological hippocampi obey the finite-capacity allocation principle remains an open empirical question.

4.3 Implications and scope

The capacity-allocation principle aligns with the invariance–disentanglement reading of the information bottleneck [Tishby et al., 1999, Achille and Soatto, 2018]: bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding (Figure **??**). The framing has a quantitative skeleton: a hidden state with effective rank d allocates dimensions among (i) directly carrying encoder-derived features (capacity d_e), (ii) integrating sensor history (capacity d_i), and (iii) carrying task-relevant non-position content (capacity d_o), subject to $d_e + d_i + d_o \lesssim d$. Bottleneck encoders force $d_e \to 0$, leaving d_i to dominate the linear position code; rich encoders enlarge d_e, so policy gradient minimises d_i down to whatever residual the integration route still needs. Our Procrustes / participation-ratio / eigenspectrum-α measurements collectively probe d (effective rank ≈ 4 for blind, lower for coarse where the encoder collapse shrinks d overall, Appendix E.5); the magnitude-axis collapse follows from d_i shrinking as d_e grows. We use IB descriptively rather than mechanistically; Saxe et al. [2019]'s critique of IB-deep-learning theory cautions against absolute MI-magnitude claims, but the cross-condition *ordering* reported here (linear collapse, MLP recovery, MINE rank in Appendix E.4) is robust to those concerns.

The capacity-allocation principle is architecture-agnostic: it predicts a content-level allocation in any system where a finite-capacity memory module reads a pretrained encoder. For transformer-based PointGoal navigators (e.g. Zeng et al. 2024; cf. Whittington et al. 2022 for the transformer/hippocampus correspondence), where "memory" lives in the KV cache or attention over rollout history, the principle predicts the same monotonic anti-correlation reported in Figure 2a: attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders. The cleanest falsification target is direct: train a transformer navigator under our 5 sensor conditions; observing the opposite pattern would falsify the principle's architecture-independence and bound it to recurrent-state systems. The same logic re-frames recent VLM spatial-reasoning failures [Ramakrishnan et al., 2025]: rich pretrained encoders may divert capacity from the downstream substrate's spatial code rather than merely fail to teach it, and bandwidth-restricting perception (foveation, log-polar) may be *cognition-enabling* rather than only compute-saving. The relevant axis is encoder spatial-feature *variety* per step rather than raw input bandwidth (Appendix G); a finer multi-point sweep within recurrent agents is left to future work.

As a partial test of the architecture-independence claim (Figure 9), we ported the encoder-bandwidth axis to a different environment, encoder family, and training regime: frozen DINOv2-Base [Oquab et al., 2024] patch features at five resolutions (mirroring our Habitat chassis: BLIND, COARSE, FOVEATED, UNIFORM, FOVEATED-LOGPOLAR) feed a small 2-layer LSTM trained on Memory Maze [Pasukonis et al., 2023] 9×9 offline trajectories with next-feature prediction. We then probe `agent_pos` from the LSTM hidden state (Ridge linear and 4-layer MLP, frame-level CV with 80/20 random split, eval window steps 250–500; full pre-registration in `scripts/probing/world_model_probe/PRE_REGISTRATION.md`). Three pre-registered conditions land cleanly. (1) Linear R^2 is monotone-increasing with encoder bandwidth: BLIND 0.01, COARSE 0.38, FOVEATED 0.41, FOVEATED-LOGPOLAR 0.41, UNIFORM 0.45. (2) Non-linear R^2 (4-layer MLP) is uniformly high in all sighted conditions (≈ 0.998) and at chance for BLIND (-0.0003). (3) The non-linear gap (MLP$-$linear) is monotone-*decreasing* with bandwidth: COARSE $+0.62$, FOVEATED $+0.59$, FOVEATED-LOGPOLAR $+0.59$, UNIFORM $+0.55$ — format-shift recovery is largest where linear readability is smallest, exactly the dichotomy our Habitat result reports. *Direction relative to Habitat.* The Memory-Maze setup deliberately omits the GPS/compass channel our Habitat agents receive; without that explicit positional anchor, the encoder is the only sensory route to absolute position, and BLIND cannot decode it (no signal, no integration anchor). The bandwidth-allocation *tradeoff* (where blind's integration route compensates) thus requires a competing route; the format-shift dynamic (linear-readable code emerges from a non-linear substrate as bandwidth grows)

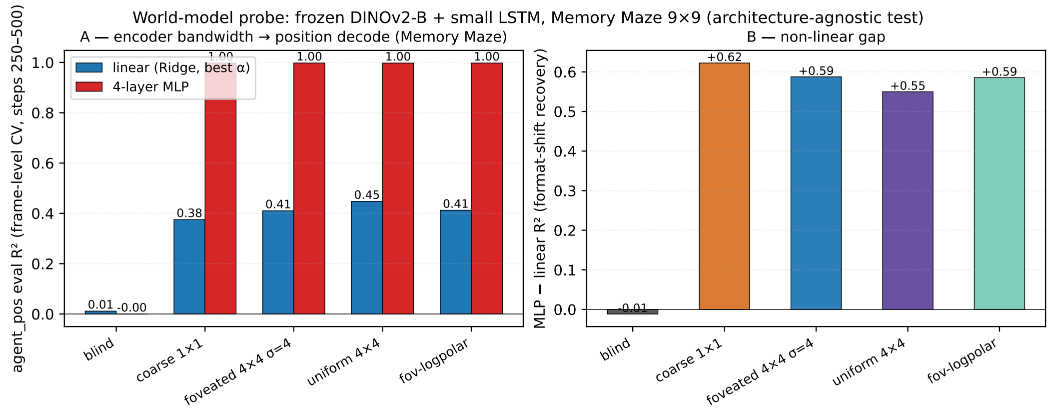


Figure 9: **Architecture-agnostic confirmation: format shift reproduces on Memory Maze with a frozen DINOv2-Base encoder feeding a small LSTM.** *(A)* Linear (Ridge, best of $\alpha \in \{0.1, 1, 10, 100, 1000\}$, blue) and 4-layer MLP (red) eval R^2 for `agent_pos` from the LSTM hidden state, frame-level random CV (80/20), eval window steps 250–500 of T=500 Memory-Maze trajectories. Linear R^2 is monotone-increasing with encoder bandwidth (BLIND $0.01 \to$ UNIFORM 0.45); MLP R^2 saturates at ≈ 0.998 for all sighted conditions and is at chance for BLIND. *(B)* The non-linear gap (MLP−linear) is monotone-*decreasing* with bandwidth: format-shift recovery is largest where linear readability is smallest (COARSE $+0.62$ vs. UNIFORM $+0.55$), the same dichotomy our Habitat agent shows in Figure **??**. The opposite linear-R^2 direction relative to Habitat reflects the absence of GPS/compass channels in the Memory-Maze setup — without an alternative positional anchor, the encoder is the only route, and BLIND carries no signal.

generalises across both regimes. The two together place the broad "encoder structure shapes memory format" claim on a different environment, encoder family, and policy regime, while clarifying that the specific direction (high-bandwidth-rich vs. bottleneck-rich) is mediated by which sensory channels carry positional information.

4.4 Limitations

(i) *Single seed per condition*, following the comparative-cognition modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]. Cross-condition rigour comes from convergent multi-method evidence (linear and MLP probing, CKA, Procrustes, subspace divergence, LOSO, lag-k, $5{\times}5$ transplant, Skaggs, excursion-forgetting all rank the conditions in the same direction — unlikely under single-seed noise alone), supplemented by an independent-retraining replication of the blind condition (Appendix A). Per-condition seed-variance error bars over a multi-seed grid are reported as future work. (ii) *Method-family and content-axis scope.* Methods we do not report and that could change a finding's strength: dynamical-systems analysis (fixed-point/slow-point structure [Sussillo and Barak, 2013]; Dynamical Similarity Analysis [Ostrow et al., 2023] as a dynamics-aware Procrustes alternative) would speak to the format axis at the level of computational primitives. We piloted DSA on our h_t time-series (10-d PCA pre-reduction, 50-step trajectories) and found the cross-condition signal sensitive to HP (n_delays, rank), rollout protocol (excursion-warmup vs. deterministic-rollout), and episode-phase alignment (blind has median episode length $\sim 2.5\times$ sighted, so first-T-step samples occupy different behavioural phases) at our explored settings; the 5-condition pattern under the cleanest protocol (deterministic-rollout, all conditions, T=50) showed sighted clustering at floor ~ 0.10 angular while blind was distant (~ 0.27), but blind's within-condition split-half DSA was also ~ 0.23, leaving the SNR insufficient to publish a definitive ranking. Information-theoretic estimators of $I(h_t; \mathrm{pos}_{t+k})$ would replace decodability R^2 with predictive-horizon nats; gaze placement and a $\sigma_{\max}$ strength sweep within the foveated condition test the rich-encoder content axis. All four are left for follow-up. (iii) *Architecture, algorithm, and task scope.* We use a recurrent (LSTM) backbone with DD-PPO, PointGoal, and a ResNet-18 visual encoder, matching the canonical pipeline of [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018] for direct comparability; the architecture choice is replication-driven rather than central to the claim, and the principle is stated at the encoder-memory interface (architecture-agnostic).

A reasonable concern is whether the encoder–memory race is an LSTM artefact (e.g. vanishing-gradient gating limitations); two within-paper checks argue against it: 2-layer MLP probes recover position in rich-encoder conditions ($R^2 \in [0.51, 0.73]$), so the LSTM *does not* discard position — the linear-readable axis tracks encoder bandwidth, a property of the linear DD-PPO action head; and the cross-condition memory transplant (§3.2) is a behavioural test outside any probe framework that reproduces the same ordering. The encoder–memory race is in principle architecture-agnostic; the cleanest falsification target is a transformer-based navigator under the same conditions, identified in §4 but not verified here. The magnitude- and format-axis patterns could also shift under different RL algorithms, different navigation tasks (ObjectGoal, ImageGoal, exploration), or different visual encoders.

5 Conclusion

A five-condition visual-sensor ablation in a recurrent PointNav agent shows sensor structure as a representational-content axis whose effects cohere under a single capacity-allocation principle measured along three axes: *magnitude* (bandwidth–allocation tradeoff between encoder and recurrent-integration routes), *format* (within-condition scene-invariance, across-condition non-interchangeable subspaces with asymmetric memory transplant, and a King–Dehaene-style temporal-format dissociation in which the blind code rotates over time while the coarse code stabilises), and *consumption* (probe-readable position dissociates from policy reliance). The bandwidth–allocation ordering is preserved on held-out MP3D, blind units integrate over a $2.1-2.5\times$ longer Murray-cortical-hierarchy timescale than every sighted condition, and the pattern is at high level consistent with independent animal-navigation findings on hippocampal compensation under sensory deprivation, cross-modality remapping, and the no-/global-remapping axis under hidden-state inference [Sanders et al., 2020] — framed as an open bio–AI hypothesis rather than a demonstrated parallel.

Methodologically, we view the paper as a step toward a *comparative cognitive neuroscience of artificial navigation agents*: a research programme in which matched-condition populations of trained agents are analysed with the same set of metrics — population geometry, intrinsic timescales, dynamical fixed points, predictive-coding residuals, persistent-homology of the state manifold, mixed-selectivity / splitter analyses, communication subspaces, transfer entropy — that is used to map biological cortex and hippocampus. We piloted this analytic suite on the present sensor axis (Figures 4, 5, 9): temporal-axis methods (TGM, Murray timescales) and topological methods (persistent homology) detect strong cross-condition signal aligned with capacity allocation, and the format-shift dynamic itself reproduces in a frozen encoder feeding a small LSTM on a different environment. Beyond this paper, the same suite can be applied to architecture-axis populations (recurrent vs. transformer world models), training-curriculum axes (sparse vs. dense reward), or biological-vs-silicon axes (LSTMs trained on egocentric video versus rodent CA1 recordings under matched stimuli). The goal is not to claim that artificial agents *are* brains, but that the analytical methods used to ask "how does this neural system represent its world?" transfer to artificial systems where the relevant variables can be held fixed — and that the cross-condition contrasts produce theoretically informative signatures (capacity allocation, format shifts, dynamical regimes) of the same kind comparative neuroscience uses to understand sensor-niche adaptation in animals. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators (the cleanest falsification target, §4). Cross-condition rigour rests on multi-method convergence and an independent-retraining blind replication (Appendix A); a multi-seed grid, finer encoder-bandwidth sweep, and gaze-placement controls are reported as future work.

Code and checkpoints. Code is available at `https://github.com/alunxu/foveated-cog-map`; the four canonical converged checkpoints plus intermediate cross-training checkpoints (used for the §3.1 substitution-dynamics figure) are hosted at `https://huggingface.co/alunxu/spatial-memory-checkpoints`.

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   Justification: The abstract and §1 state the three hypotheses H1/H2/H3 with explicit signatures (what pattern supports each), and preview the three findings with specific magnitudes (path-history R^2=0.56, CKA<0.004, compass R^2=0.94 vs GPS R^2=0.03). These are expanded in §4 and §5 with no stronger claims than the data support.

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   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §5.4 explicitly lists six limitations (i–vi) covering checkpoint staleness, single-seed H3, missing proposed analyses, foveation-model simplification, linear-probes-only, and LSTM-architecture scope.

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   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
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   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, Hewitt–Liang control). Appendix A gives the ablation's loss formulation and coefficient.

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   Justification: We report single-seed numbers per condition. This is documented as Limitation (ii) in §5.4. Multi-seed replication for tighter confidence intervals is in the immediate follow-up.

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   Justification: §3.7 reports per-condition training budget (5–7 days on 2×V100 for 250M frames) and probing budget (<1h per condition on one V100).

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A Reproducibility: hyperparameters and implementation notes

Training hyperparameters (DD-PPO, all five conditions identical) 4-GPU `torchrun`; `num_steps=256`, learning rate `lr=2.5e-4` with `lr_decay=False`; `num_environments=16/worker`; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from Wijmans et al. [2020].

Architectural details (sensor encoding) Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (`nn.Embedding` with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the Wijmans et al. [2020] Wijmans-default sensor stack.

Implementation notes (foveated condition) The foveated condition disables the running-mean-and-variance input normaliser (`RunningMeanAndVar`) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected; the ablation is in the codebase under `src/habitat/foveated_normalised_policy.py` (normaliser *enabled*) for future work that wants to revisit this choice.

Implementation notes (numerical stability) Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) gradient sanitisation in `wijmans_policy._safe_before_step` replaces NaN/Inf gradients with zero before each PPO update; (ii) a forward-looking `geodesic_distance` clamp in `nan_safe_measures.py` prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories.

Log-polar foveation transform The log-polar control (Appendix G) uses `LogPolarFoveationTransform` in `src/habitat/torch_foveation.py`: resample the 128×128 avg-pooled image onto a 64×64 ($n_\rho \times n_\theta$) log-polar grid before the ResNet-18 backbone. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

B Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

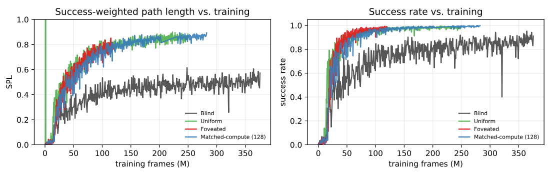
Length-matching ablation. Under deterministic rollouts, episode lengths span ∼100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the H1 ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe. The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

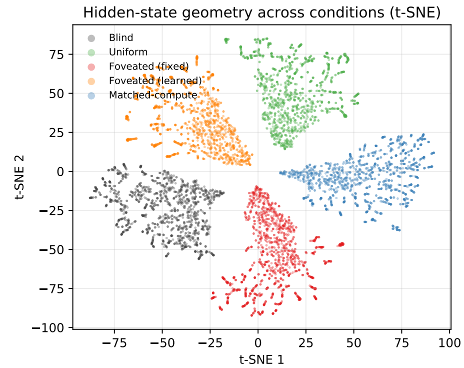
C Probing details (Layer 0 sensor branch and MLP probe)

LSTM Layer 0 sensor branch. The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely.

MLP probe details. For each condition we fit a 2-layer MLP (hidden dim 256, ReLU, L_2 weight decay 10^{-4}) on the same top-layer hidden states and GPS labels as the linear Ridge probe, with the same



(a) Training curves (TensorBoard event-file scalars). All three sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all four conditions (perplexity 30). The four conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§3.2).

Figure 10: Supplementary visualisations referenced from the main text.

5-fold episode-level CV. Top-layer recovery: foveated $R^2 = 0.51$, uniform 0.55 (vs. Ridge's chance R²). The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the log-polar foveation control (Appendix G) targets the encoder-spatial-output axis at one intermediate point.

D Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); the encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all four conditions ($R^2 \geq 0.81$), and the real-vs-permuted-label probe gap (Hewitt–Liang selectivity) tracks GPS R^2; the chance-level rich-encoder GPS R^2 is therefore a hidden-state property, not a probe artefact.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix B.

Linear-probe artefacts. Two complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the 5×5 matrix's asymmetry adds structure no linear-similarity measure can read. *(b) Non-linear probe:* a 2-layer MLP recovers $R^2 \in [0.51, 0.73]$ from rich-encoder top-layer states (Appendix C). H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.

E Supplementary visualisations

E.1 Capacity-allocation: additional geometric tests

The three figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-allocation (Fig. **??**) and LOSO scene-invariance (Fig. **??**) results in §3.1.

E.2 Single-direction β-scrubbing reveals redundant population coding

The Ridge probe identifies a 2-d direction β in $\mathbf{h}_2$ that linearly predicts (x, y). We test whether this direction is *causally responsible* for the recoverable position code by projecting $\mathbf{h}_2$ onto the null-space

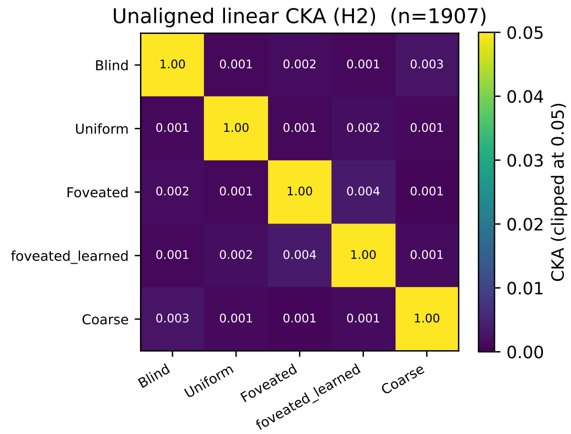


Figure 11: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. Within-condition split-half CKA ≈ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure **??**); CKA is the least informative of the three H2 measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

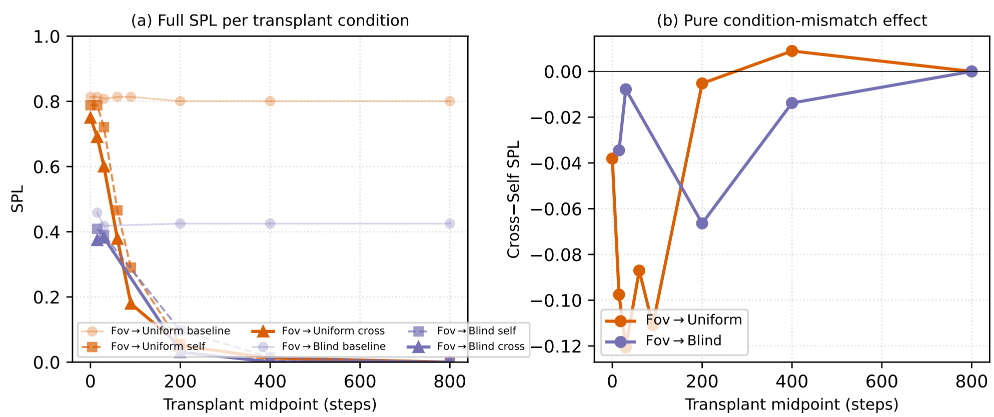


Figure 12: Memory-transplant midpoint sweep on two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps (n=100–150 episodes/cell). The full $5{\times}5$ matrix in Figure **??** (right) is at midpoint t=30. *(a):* full SPL per condition — baseline / self / cross. The t=0 self/cross collapse to a protocol-noise floor (so the diagonal in Figure **??** is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. *(b):* cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at t=30 in Figure **??** is therefore neither a t=30-specific artefact nor a late-midpoint inflation.

of β (rank-510 subspace) and re-evaluating both Ridge and MLP probes (Heimersheim & Nanda 2024 noising-style patching: tests whether β was *necessary* for the encoded behavior [Heimersheim and Nanda, 2024]).

	Linear orig	Linear scrub	MLP orig	MLP scrub
Blind	+0.95	+0.93	+0.94	+0.95
Coarse	+0.72	+0.62	+0.79	+0.84
Uniform	−1.08	−1.46	+0.42	+0.31

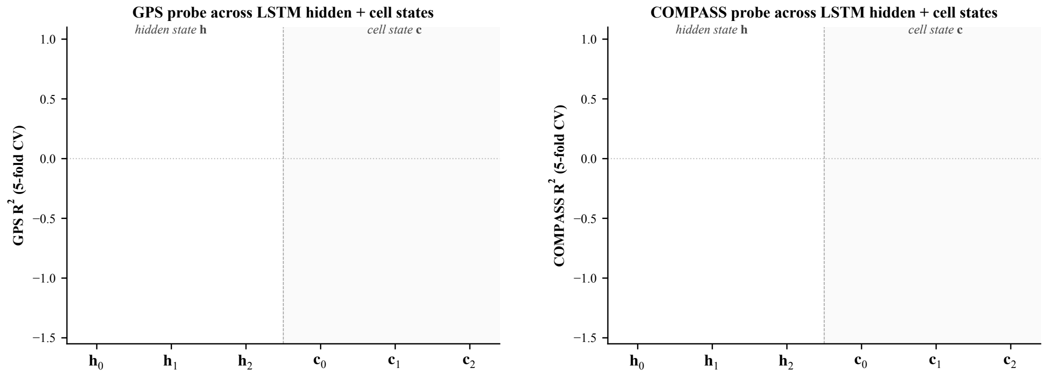


Figure 13: Linear probe (R^2, 5-fold CV) at each LSTM hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left:* GPS. *Right:* compass. Three patterns. *(i) At* $\mathbf{h}_0$ *all conditions decode GPS at* $R^2 \sim 0.95$ (the GPS sensor embedding enters the LSTM concat there, regardless of vision condition). *(ii)* $\mathbf{h}_1$ *dips for every condition* including blind ($R^2 = +0.22$); information passes through middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for bottleneck. *(iii) Cell states* $\mathbf{c}_\ell$ *do* not *carry the GPS code as cleanly as* $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The H1 claim about "top-layer linearly-decodable GPS" is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.

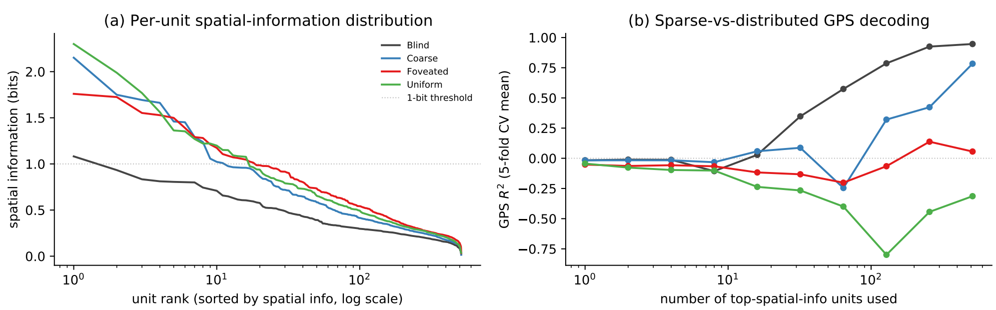


Figure 14: Population coding analysis (referenced from §**??**). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information* $\neq$ *position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

The MLP probe is essentially unaffected by scrubbing across all three conditions ($\Delta R^2 \in [-0.05, +0.11]$). The Ridge probe shows small drops for Blind (-0.02) and Coarse (-0.10), and a more-negative shift for Uniform (the absolute value increases due to the variance of negative R^2 estimates). Two interpretations: (i) the rank-2 subspace identified by Ridge is one of many redundant directions encoding position — when removed, Ridge re-fits to a different direction in the remaining 510 dims; (ii) MLP, having access to non-linear combinations of remaining dims, trivially recovers the position code from this redundant population. This is the population-coding-redundancy property familiar from biological place-cell systems [Stringer et al., 2019]: single-cell or low-rank ablation does not break the code because of high-dimensional distributed representation. Methodologically, single-direction scrubbing is too weak to causally localise position information in $\mathbf{h}_2$; higher-rank

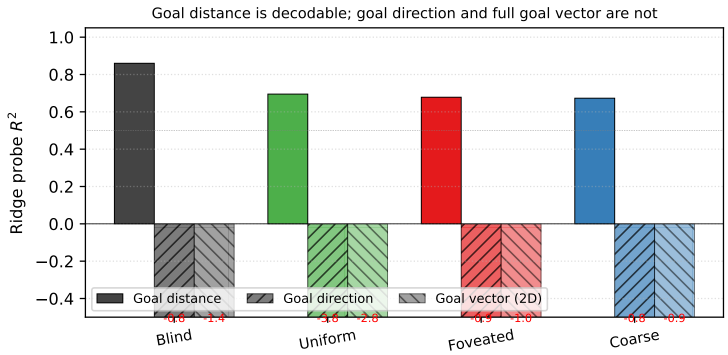


Figure 15: Goal-vector probe R^2 (referenced from §**??**). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

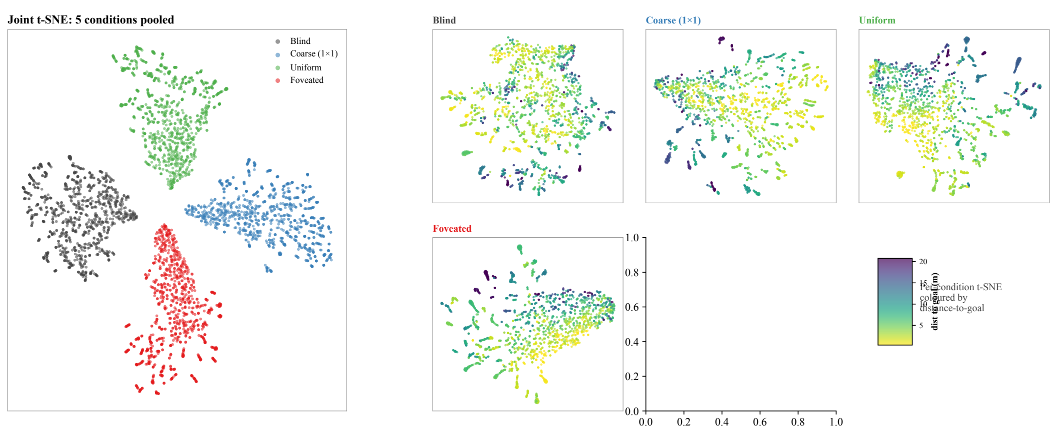


Figure 16: t-SNE of top-layer LSTM hidden states $\mathbf{h}_2$, supplementing the linear-similarity H2 results (§3.2). *Left:* joint t-SNE of 5 000 samples (1 000 per condition) in 512-d hidden space. The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right:* per-condition t-SNE coloured by distance-to-goal ($n = 2\,000$ per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with the H1 finding that blind's hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

ablation, attribution-based pruning, or path-patching [Heimersheim and Nanda, 2024] would be next steps. Foveated is deferred to the 250M re-probe.

E.3 Predictive horizon: does h_t encode future positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k}. We test the analog claim for our recurrent state: does a linear or MLP probe from $\mathbf{h}_t$ recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

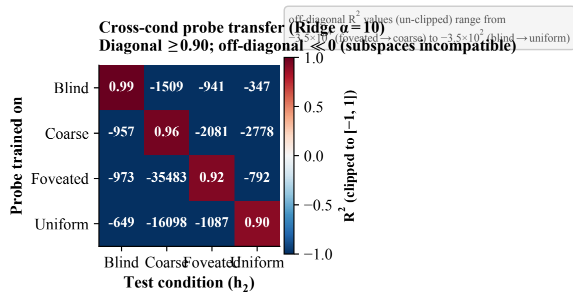


Figure 17: **Cross-condition probe-transfer matrix.** Ridge-probe trained on condition A's top-layer states, applied to condition B's. Diagonal (own-probe) $R^2 \in [0.90, 0.99]$. Off-diagonal R^2 is *catastrophically negative* (-3.5×10^2 to -3.5×10^4 unclipped; clipped to $[-1, 1]$ for visualisation): the position-axes encoded across conditions are not just orthogonal in subspace orientation (Figure 28) but jointly incompatible at the readout level. Even when the same scenes are visited and the same task is solved, no single linear position-direction transfers between conditions.

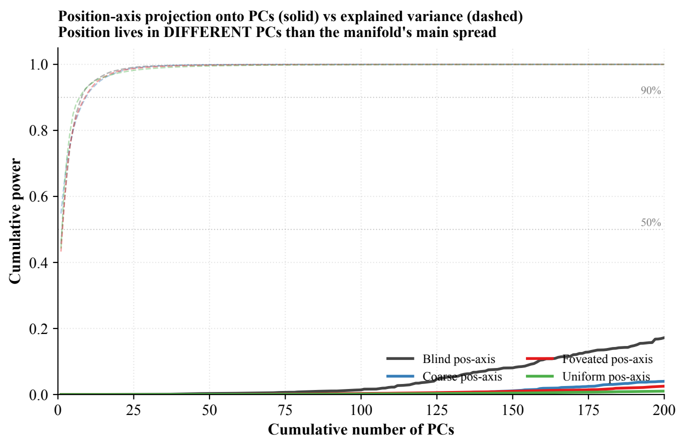


Figure 18: **Position-axis lives in low-variance directions of $\mathbf{h}_2$.** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold's own variance distribution). For all four conditions, the manifold's variance is concentrated in $\sim$ 7–9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of $\mathbf{h}_2$ across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure **??**).

	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 50$
Linear (Ridge $\alpha = 10$) R^2, all-episode pooling, episode-level 5-fold CV						
Blind	$+0.94$	$+0.94$	$+0.94$	$+0.92$	$+0.90$	$+0.79$
Coarse	$+0.57$	$+0.57$	$+0.57$	$+0.56$	$+0.50$	$+0.37$
Foveated	$+0.06$	$+0.06$	$+0.08$	$+0.13$	$+0.19$	$+0.06$
Foveated-logpolar	$+0.17$	$+0.18$	$+0.17$	$+0.11$	-0.14	-0.50
Uniform	-4.31	-4.32	-4.32	-4.26	-4.39	-10.14

Three observations. *(i) Blind exhibits a long predictive horizon.* Linear R^2 from $\mathbf{h}_t$ to $(x, y)_{t+k}$ remains ≥ 0.90 from $k = 0$ to $k = 20$ (only 0.04-point drop), and 0.79 at $k = 50$. That is, the same linear projection that decodes *current* position from $\mathbf{h}_t$ at $R^2 = 0.94$ also decodes positions 20 steps in the future at $R^2 = 0.90$. This is a strong form of the predictive-map signature predicted by SR theory: $\mathbf{h}_t$ encodes future-state occupancy, not just current state. *(ii) Coarse follows the same pattern at lower magnitude* (linear $R^2 \in [0.50, 0.57]$ across $k = 0$–20, dropping to $+0.37$ at $k = 50$).

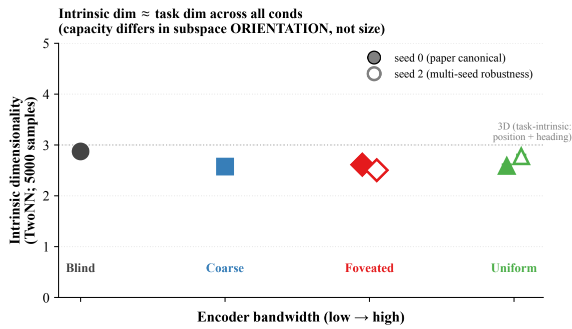


Figure 19: **Intrinsic dimensionality (TwoNN).** Facco et al. [2017]-style estimate of the intrinsic dimensionality of $\mathbf{h}_2$, on 5 000 samples per condition (paper-canonical seed-0 filled markers; seed-2 robustness check hollow). All four conditions yield ID ≈ 2.5–2.9, comparable to the task-intrinsic dimensionality (2-D position + 1-D heading $\approx$ 3). The same intrinsic dim across conditions, combined with strongly differing linear R^2 and orthogonal subspaces (Figure 28), shows capacity-allocation is about the *orientation* of the task-shaped manifold within the 512-d ambient space rather than about its size.

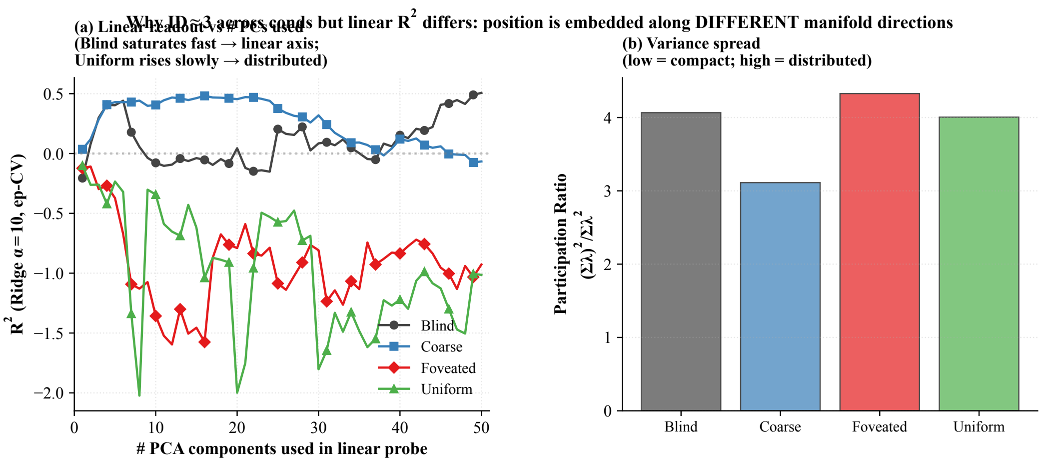


Figure 20: **Linear readout vs. # PCs.** *(a)* Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with ≤ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). *(b)* Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 18).

(iii) Rich-encoder conditions have no clear predictive horizon. Foveated stays in $[0.06, 0.19]$ across $k = 0$–50 with no monotonic decay — the small positive values reflect probe-level noise rather than a sustained future-state code. Foveated-logpolar starts at $+0.17$ but turns negative by $k = 20$. Uniform is already strongly negative at $k = 0$ ($R^2 = -4.31$) and grows more negative. Rich-encoder $\mathbf{h}_t$ encodes neither current nor future position linearly — consistent with our LOSO scene-invariance reading that rich-encoder position codes are scene-conditional and only non-linearly recoverable, but adds the temporal axis: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation.

(See main Figure 2c for the predictive-horizon panel; this appendix expands on its theoretical context.)

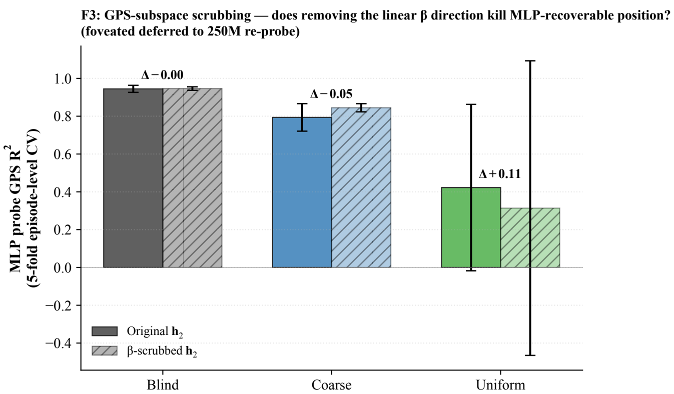


Figure 21: β**-subspace scrubbing.** For each cond (foveated deferred), Ridge β identifies a 2-d direction in $\mathbf{h}_2$ that linearly predicts (x, y); scrubbed $\mathbf{h}_2 = (I - \beta^+\beta)\,\mathbf{h}_2$ removes this 2-d subspace. MLP probe is essentially unaffected by scrubbing across all three conditions: position information is redundantly distributed across many h_2 dimensions, not concentrated in the 2-d β subspace.

E.4 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in §3.1 on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(\mathbf{h}_2; \mathrm{pos})$ per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X;Z) \geq \sup_T \mathbb{E}_{P_{XZ}}[T(x,z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x,z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired $(\mathbf{h}_2, \mathrm{pos})$ along episodes; marginal samples are obtained by within-batch shuffling of z. We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4}, $4{,}000$ training steps per condition, and the bias-corrected gradient (eq. 12 of Belghazi et al. 2018) to mitigate the SGD bias of the log-denominator estimate.

E.5 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of Stringer et al. [2019] to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All five conditions show a tight power-law eigenspectrum (Figure 23, $R^2 \geq 0.99$). The fitted exponent shows a sharp bottleneck-vs-rich-encoder split:

Condition	α	R^2	Var. top 10 PCs
Blind	3.85 ± 0.03	0.99	95%
Coarse	1.80 ± 0.01	1.00	82%
Foveated	1.72 ± 0.01	1.00	76%
Foveated-logpolar	1.80 ± 0.01	1.00	95%
Uniform	1.90 ± 0.01	1.00	91%

Two observations. (i) Blind sits well above the critical exponent $1 + 2/d_{\mathrm{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\mathrm{task}} = 3$ stimulus (current position + heading), and far above Stringer's V1 value ($\alpha \approx 1.04$). With $\alpha = 3.85$, blind's eigenspectrum decays sharply: variance is concentrated in the top few PCs (top-10 already capture 95%). We do not claim that Stringer's specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d-dim stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) The four sighted conditions cluster tightly around $\alpha \approx 1.7$–1.9, all near the $1 + 2/d_{\mathrm{task}} \approx 1.67$ critical exponent — the smoothness/efficiency border. Blind is the outlier above, in a few-dim concentrated regime, while the four sighted conditions distribute variance across more PCs. The split

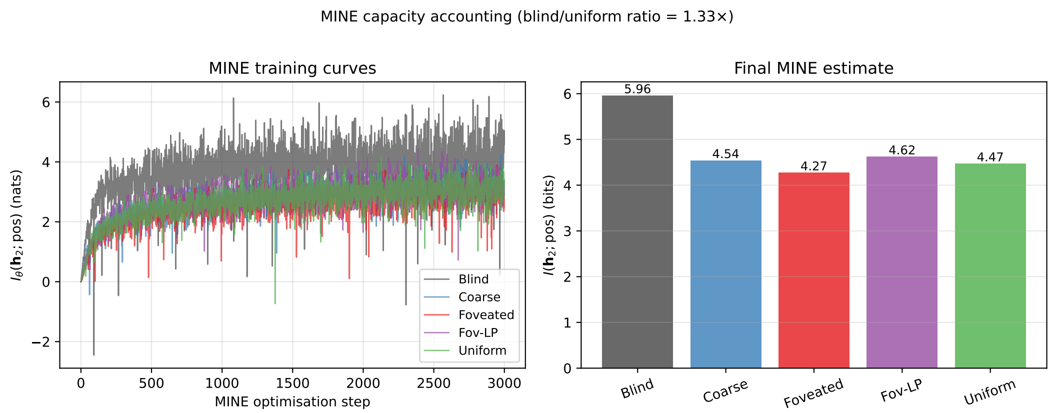


Figure 22: **MINE capacity accounting.** *(a)* MINE training curves: $I_\theta(\mathbf{h}_2; \text{pos})$ in nats vs. optimisation step, per condition. All curves converge by $\sim 3{,}000$ steps. *(b)* Final $I(\mathbf{h}_2; \text{pos})$ in bits per condition (mean $\pm$ std across 3 MINE seeds): Blind 6.01 ± 0.06, Coarse 4.55 ± 0.03, Foveated 4.30 ± 0.04, Foveated-logpolar 4.66 ± 0.04, Uniform 4.47 ± 0.06. Total mutual information decreases with encoder bandwidth, but the decrease is gentle ($\approx 1.3\times$ blind/uniform) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The within-sighted ordering (foveated $<$ uniform $<$ coarse $<$ foveated-logpolar) is reproducible across MINE seeds (per-cond std ≤ 0.06 bits, much smaller than the within-sighted gap ≈ 0.36); the dominant signal is the bottleneck-vs-rich-encoder gap (Blind 6.01 vs sighted mean 4.49). The interpretation: rich-encoder conditions retain substantial position information in $\mathbf{h}_2$, but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure **??**).

Table 2: **Geometric scalars: per-condition manifold shape (companion to Figure 23; referenced from §3.2).** Participation ratio (PR), intrinsic dimension (ID-MLE / ID-CDF), and eigenspectrum power-law exponent α on $\mathbf{h}_2$. All values from post-retrain hp-consistent agents (3-seed mean over 30,000-step subsamples; std < 0.05). Blind separates on α (sharper tail) and on ID (lowest); the four sighted conditions cluster on α and ID (1.73–1.92, 2.04–2.41). PR varies more across sighted conditions and does not track encoder bandwidth monotonically; we read it as different geometric distributions of the visual route's residual variance rather than a clean ordering.

Condition	PR	ID-MLE	ID-CDF	α
Blind	4.41	**1.72**	**1.54**	**3.63**
Coarse	5.25	2.41	2.20	1.81
Foveated-logpolar	1.70	2.25	2.04	1.88
Foveated	9.11	2.41	2.30	1.73
Uniform	3.09	2.32	2.17	1.92

bottleneck-vs-rich-encoder reading: blind's heavy reliance on the integrated GPS code crystallises into a low-dimensional eigenspectrum, while sighted conditions (including coarse, whose 1×1 encoder still feeds a ResNet-18 stack with diverse channels) distribute hidden-state variance across more directions to support encoder-route information.

E.6 Persistent-memory failure-mode catalog

The main-text Figure 29 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 24) shows four cases per condition spanning the per-condition margin range (most-tries-new $\rightarrow$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5\,\text{m}$ (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

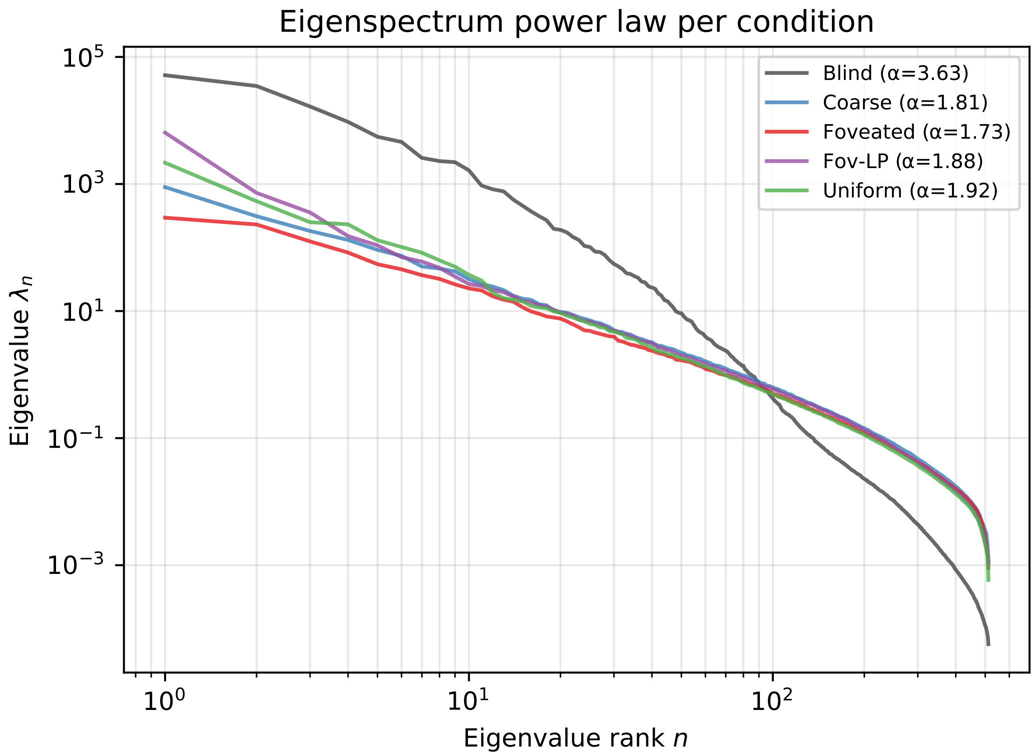


Figure 23: **Eigenspectrum power-law per condition.** *(a)* Eigenvalue λ_n vs. PC index n for $\mathbf{h}_2$, log–log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. *(b)* Fitted power-law exponent α per condition (linear fit on log-λ_n vs. log-n, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-4, unified hp)
Encoder feature-map probe	where position information is held	**landed; see §4**

F Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from the main paper. All four are submitted.

Foveation strength sweep (F1–F4) — planned for follow-up, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 4×4 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output

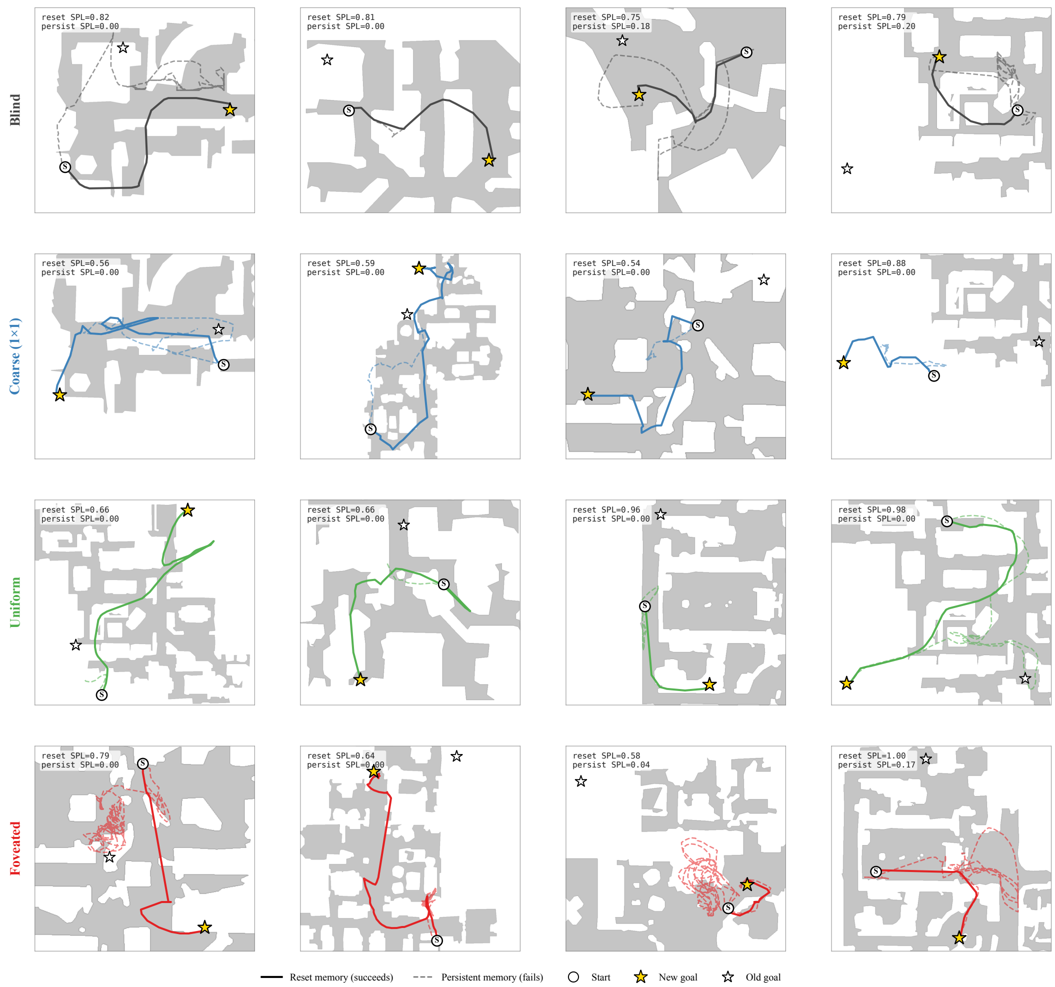
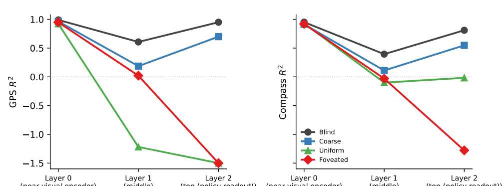
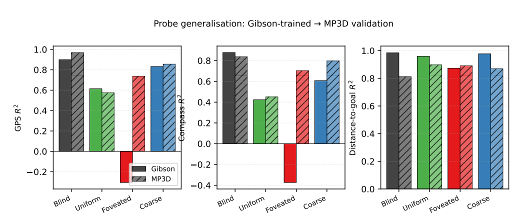


Figure 24: Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution. Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at "tries-new" (most cases attempt new goal direction but fail to reach); Uniform clusters at "locks-onto-old" (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single seed per condition (cogsci modelling convention; see §4.4).



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all four conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 26a (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-rich-encoder partition is the same.

Figure 25: Per-layer and MP3D companion figures (referenced from §3.1).

drops to $\sim 2\times2$, much closer to coarse's 1×1 than to uniform's 4×4. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*.

The preregistered prediction (written before result was available) was that log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.58$ and uniform's ≈ 0, with corresponding behavioural shifts; if log-polar matched uniform, the encoder-spatial-output-dimensionality mechanism would be wrong. The result (next paragraph) falsifies that simple reading.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper's interpretation.

G Encoder-capacity scaling: log-polar foveation control

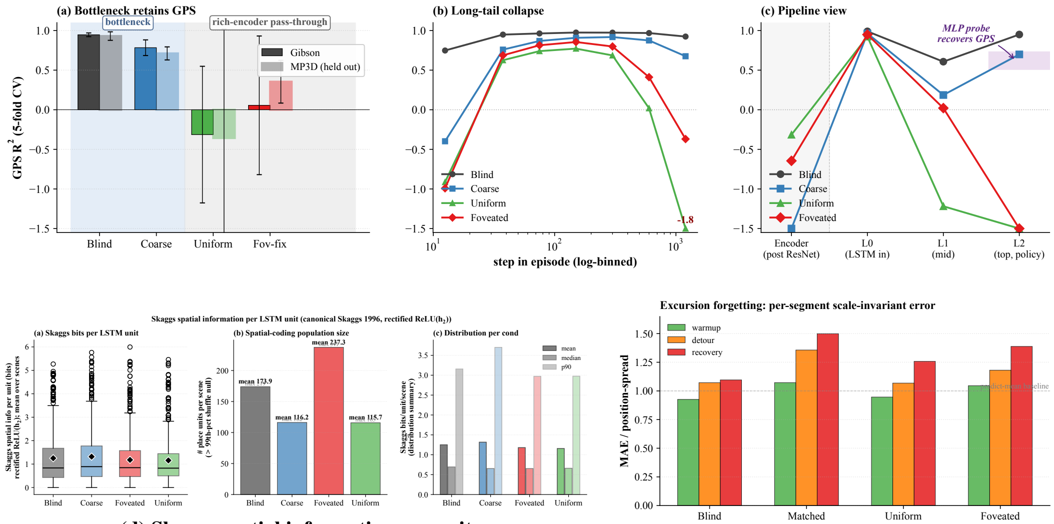
A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2\times2$, intermediate between coarse's 1×1 and uniform's 4×4 (verified by direct forward pass through the trained encoder). Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered prediction:** if the H1 mechanism is the simple "encoder spatial-output dimensionality" reading, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse's $+0.58$ than to uniform's ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per step, or another axis. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-49, ~ 245M frames) yields top-layer GPS $R^2 = -0.029 \pm 0.759$ (5-fold CV, $n = 500$ episodes) — *not* in the bottleneck regime, but indistinguishable from foveated ($R^2 = +0.178 \pm 0.659$) and clearly far from coarse ($R^2 = +0.582 \pm 0.099$). The simple "encoder spatial-output dimensionality" framing is therefore *not* sufficient to explain the H1 phenomenon: log-polar's 2×2 output does not place it in the bottleneck regime. We read the result as supporting the alternative "encoder spatial-feature *variety* per step" framing: log-polar's radial sampling preserves rich gaze-centred feature variety per timestep even at low spatial output, so the policy can still source position from the visual route and the LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure 2b) corroborates this reading: log-polar starts at $R^2 = +0.92$ at ~ 50M frames (matching all 4 main conditions' early high R^2), then decays along the rich-encoder trajectory rather than plateauing in the bottleneck range. A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at ~ 250M frames each) was outside this submission's compute window.

H Additional representational and behavioural diagnostics

This appendix collects companion figures that extend results reported numerically in the main text. They are not load-bearing for any main-text claim and may be skipped on first reading.

H.1 Magnitude axis: corroborating diagnostics

The main-text Figure 2 (§3.1) summarises the magnitude-axis story across three independent axes; this single composite figure collects the corroborating diagnostics referenced from that section in one place.



(d) Skaggs spatial information per unit

(e) Excursion forgetting

Figure 26: **Magnitude axis: corroborating diagnostics (referenced from §3.1 and Table 1).** *(a)* Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; the MP3D shift is single-seed and awaits replication). *(b)* GPS R^2 binned by step-in-episode (7 log-spaced bins): rich-encoder readability holds in the mid-episode window then collapses in the long tail. *(c)* Pipeline view: GPS R^2 along Encoder → L0 (LSTM input) → L1 → L2 (single-fit; the 5-fold CV version is condensed in main Figure 2). Readability is comparable across conditions at the LSTM input and only diverges at L2; the shaded band at L2 is the 2-layer MLP probe recovery range. *(d)* Per-unit Skaggs spatial information [Skaggs et al., 1996] is condition-flat at the headline metric; the per-unit regression analysis in §3.1 (trajectory-phase vs. raw-position R^2) is the more diagnostic reading. *(e)* Mid-episode excursion forgetting (25-step random-action perturbation; per-segment scale-invariant MAE): blind fragments most, consistent with the strongest reliance on the integrated action stream.

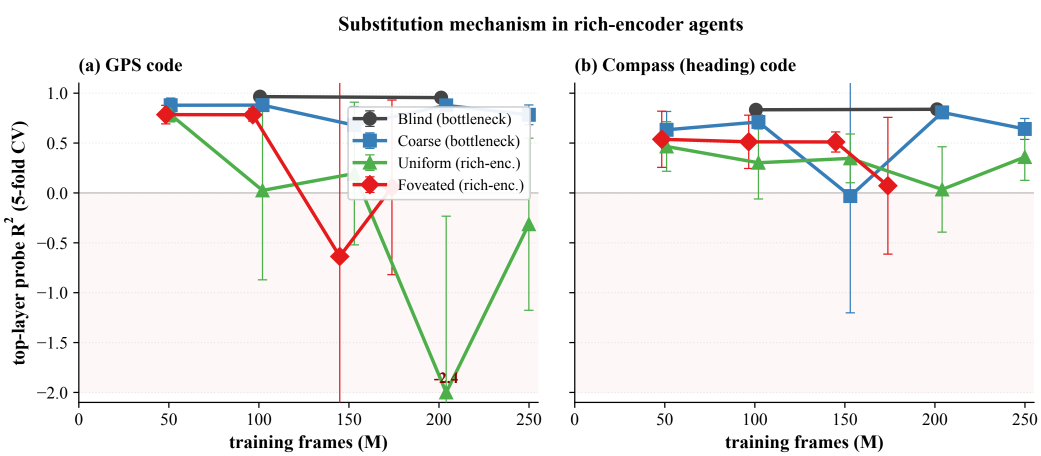
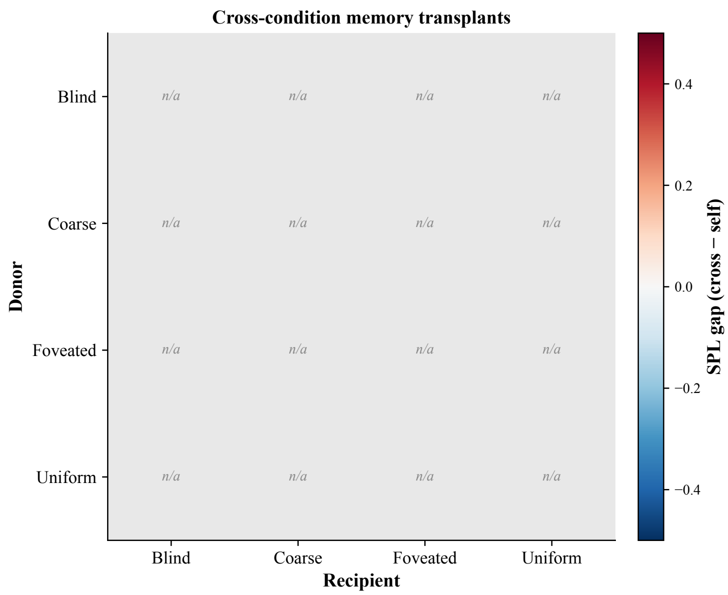
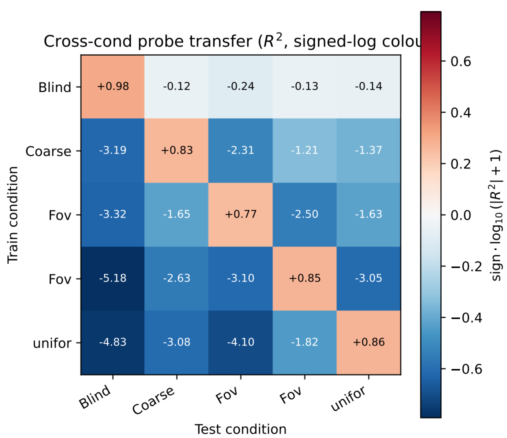


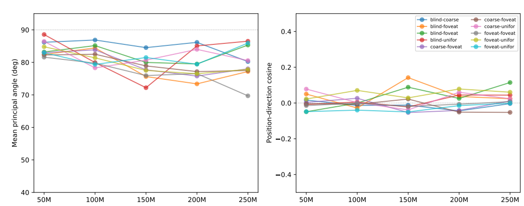
Figure 27: **Cross-training substitution dynamics: GPS and compass (companion to main Figure 2b).** Probe R^2 across training checkpoints, truncated at 250M frames. *(a)* Top-layer GPS R^2 (the source panel for Figure 2b). *(b)* Top-layer compass R^2 tells the same story at smaller magnitude — substitution applies to spatial codes broadly, not just position.



(a) $5{\times}5$ **memory-transplant matrix (full)**



(b) Cross-condition probe transfer



(c) Subspace evolution over training

Figure 28: **Format axis: corroborating diagnostics (referenced from §3.2).** *(a)* The full $5{\times}5$ memory-transplant matrix at midpoint $t{=}30$ (the per-recipient mean enters main Figure 3b as the "transplant cost" axis); the matrix shows the same direction-asymmetry condensed there. *(b)* Cross-condition probe transfer matrix (train on row, test on column): off-diagonal R^2 is catastrophically negative — the position-axes encoded across conditions are jointly incompatible at the readout level. *(c)* Subspace divergence is established early: principal angles are already near-orthogonal at the earliest probed checkpoint ($\sim$ 50M frames) and stay so throughout, indicating the structural separation is laid down well before convergence.

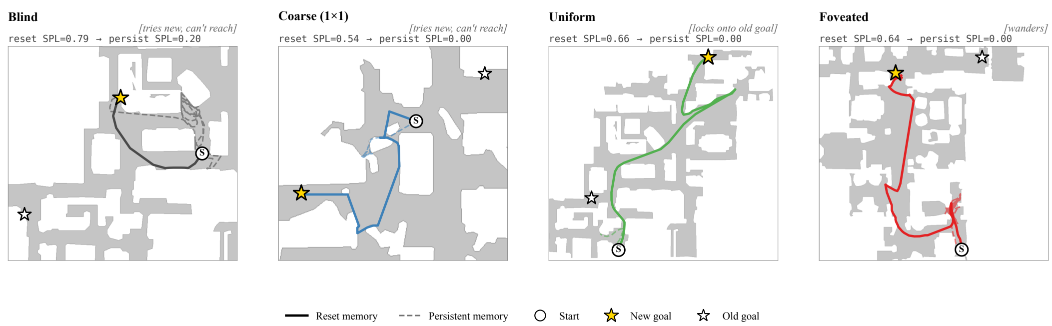


Figure 29: **Persistent-memory failure modes (referenced from §3.4): canonical paired-episode case per condition.** Extended catalog: Figure 24. Aggregate per-condition margin = avg dist(persistent end → old goal) − avg dist(persistent end → new goal) over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5$m): Blind -0.38m (n=27), Coarse -0.57m (n=35), Foveated -0.59m (n=16), **Uniform** $+1.83$**m (**n=46**, closer to OLD goal)**. Single seed.